Main Title

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COURSE

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March 29, 2026

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Contents

1 Introduction

1.1 Expectations

At the beginning of this project, we explored several concepts for an electric vehicle, aiming to go beyond the standard passenger car. We initially considered a performance-oriented Formula E platform as well as a high-efficiency commuting car. However, since individual commuting is fundamentally inefficient as a concept, we looked for more impactful solutions. Rail transport, for instance, is significantly more efficient per ton-kilometer. This led us to combine our interests in performance and efficiency by investigating the electrification of heavy-haul freight trains.
Specifically, the famous Mauritanian Iron Ore Train, which transports ore from inland mines to the coast. This project is particularly compelling because of the theoretical abundance of solar energy in the region. However, the Sahara's harsh desert environment presents a major engineering challenge for electrification. Currently, the train relies on diesel-electric locomotives operating in a series-hybrid configuration [**?**]. In 2023, the emissions from this operation exceeded 75,000 tons of CO2 equivalents (CO2e), which is roughly equal to the annual carbon footprint of 7,500 average EU households [**? ?**].
After having the idea, further research also showed that we are not the first to explore this potential. During the 2023 United Nations Climate Change Conference (COP28), a memorandum was signed to explore electrifying the rail corridor using green hydrogen [**?**], which lead us to the conclusion that there is a real use case for this project.
In contrast to most vehicle design projects, presented in the course, this project aims to design a vehicle for one specific usecase on one specific track. Although it can function as a role model for the electrification of heavy duty trains in remote and arid regions, its parameters are designed and optimized for its specific track, mass and usecase. The expectation was therefore not to create a vehicle that could be used and tested in different driving cycles, but to create an electrified version of the train and the necessary infrastructure, specifically designed for the driving cycle of transporting the iron ore from the mines to the sea, and the empty train back. Thus the design process started, by looking at the current trains specifications and its track.

1.2 Vehicle Specifications

In **??**, key numbers for the currently operating train are summarized. It has to be noted that during the research only little information about quantitative vehicle specifications could be found, which left room for some interpretation and educated guesses, e.g. by comparing other heavy duty trains.

Table 1.1: Vehicle specifications of the iron ore train [?].

Purpose	Transport of Iron Ore
Length	2500 m
Weight	5000 t (empty) - 22000 t (loaded)
Top Speed	65 km/h (empty) - 45 km/h (loaded)
Max. Gradient	0.36%
Acceleration	below 0.1 m/s^2
Track Length	700 km

The current train is operated in several configurations, typically consisting of four diesel locomotives pulling between 200 and 220 ore wagons, along with two additional carriages for mining staff. The power requirements for the operation can be divided into two distinct segments. On the outbound journey to the mines, the train is empty but must overcome a steady positive gradient, as the mines sit at an elevation of approximately 390 m above sea level [**?**].
On the return journey to the port, the situation is reversed. Although the train is significantly heavier due to the ore load, much of the route follows a negative gradient, which reduces the net energy required for propulsion. However, the "main crux" of the entire track is a relatively steep uphill section located shortly after leaving the mines toward the port **??**. This specific segment is a critical design parameter, as it determines the steady-state peak power the locomotives must provide to move the fully loaded mass. Using google earth elevation data [**?**], a gradient map of the track was extracted. This profile served as the basis for determining the train's specific driving cycle in the following sections.

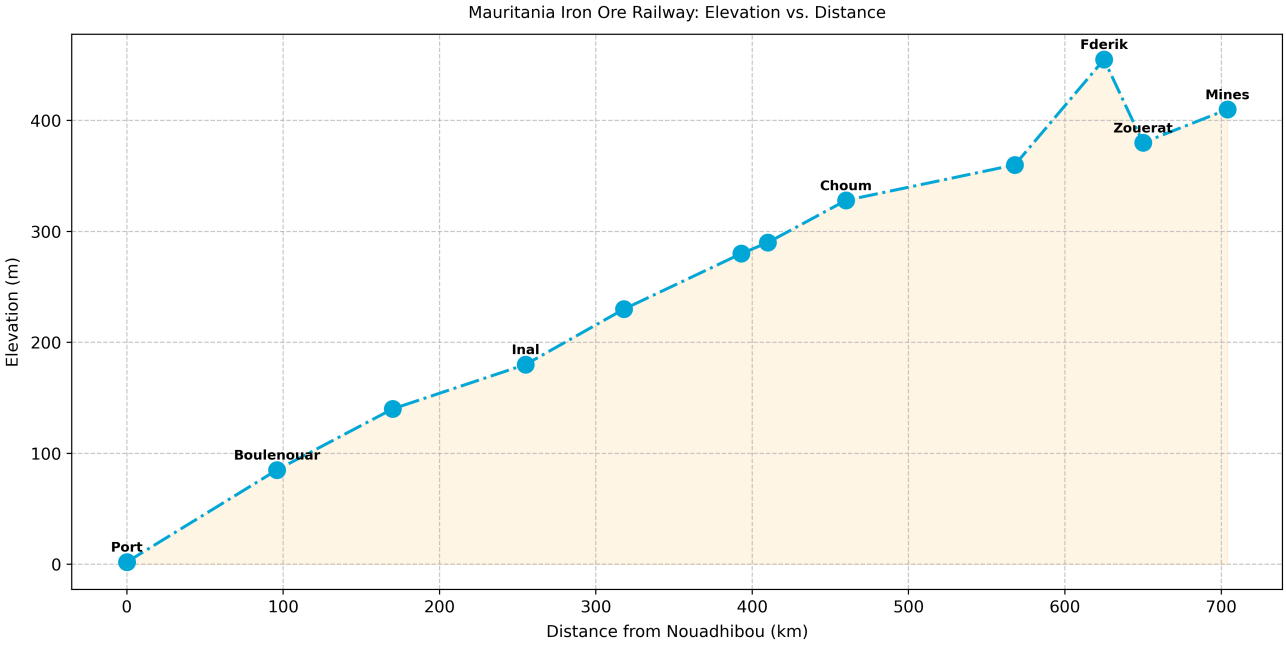

Figure 1.1: The height map of the track, extracted from google earth data of the elevation along the route [?].

Along its journey the train passes through five larger villages, where the train is reported to stop for the loading of small goods and occasionally people [**?**]. These stops, together with the start and end stations at the mines and the sea, are also depicted in **??**.

2 Design of the Driving Cycle

Given the limited quantitive information about the train itself and its velocity during the journey, the driving cycle itself became part of the design process. It had to be designed how fast the train accelerates, its speed on different sections of the track, and how long the train stops. This section explains how the track sections were modelled and the driving cycle designed, based on the height profile and stops in **??**. In order to reasonably limit the design freedom, the goal was set for the train to finish the enitre journey in under $20\,\mathrm{h}$ and $15\,\mathrm{h}$ for the west- and eastbound case, respectively. To not endanger machanical stability of the track rails, the top speeds of $45\,\mathrm{km\,h^{-1}}$ and $65\,\mathrm{km\,h^{-1}}$ were

also kept as a hard design requirement. These values are all in approximate agreement with the current train journey.
From there on, calculations for the required traction power P_t of the train for different scenarios, such as zero-gradient top speed, different accelerations and the incline part for the loaded train, were performed using

$$P_t = V\left(Mgf_r\cos\alpha + Mg\sin\alpha + \frac{1}{2}\rho A_f C_D(V)^2 + M\delta\frac{dV}{dt}\right) \tag{1}$$

in order to find the maximum value for P_t. The parameters used for these calculations are listed in **??**. It has to be noted, that **??** assumes zero wind speed, to simplify the areodynamic power

Table 2.1: Parameters used for the traction power calculations of the train.

Parameter	Value	Description
V	max. $65\,\mathrm{km\,h^{-1}}$ (unloaded), $45\,\mathrm{km\,h^{-1}}$ (loaded)	maximum velocity of the train
M	$5\times10^6\,\mathrm{kg}$ (unloaded), $22\times10^6\,\mathrm{kg}$ (loaded)	mass of the train
g	$9.81\,\mathrm{m\,s^{-2}}$	gravitational acceleration
f_r	0.0015	rolling resistance coefficient
α	max. $0.0036\,\mathrm{rad}$ (loaded), $0.0015\,\mathrm{rad}$ (unloaded)	(maximum) incline angle
A_f	$12\,\mathrm{m}^2$	aerodynamic frontal train area
c_D	1	aerodynamic drag coefficient
δ	1.02	mass factor due to inertia

loss estimations (in **??** it is shown that aerodynamic forces are relatively small anyways). Further, the maximum incline angle is small enough for the approximation of $\cos\alpha \approx 1$ and $\sin\alpha \approx \alpha$ to be used in **??**. The resulting values for P_t were then compared to each other to identify the section that requires the most power and to evaluate, if it makes sense to reduce this power, e.g. by lowering the speed or acceleration, in order to avoid oversizing of the powertrain for one section.
For simplicity of the model, it was assumed that the train accelerates to a constant speed in between stops and only slows down to come to a halt at the next village. This leaves the question of the target speed in between sections and the acceleration profile open, to fully define the drive cycle.

2.1 Steady state speed design

In the most simple version of the driving cycle, the target speed would always be equal to the maximum possible top speed after **??**. From a power perspective, the most demanding section in steady state ($\frac{dV}{dt} = 0$) is the one with the maximum incline angle α_{max}. Thus, steady state power calculations for the maximum incline on both the loaded and unloaded journey were done and depicted in **??** for the respective top speeds. As expected it can be seen that the maximum traction power, is required on the incline part of the loaded journey. Since it exceeds $13.6\,\mathrm{MW}$ for a speed of $45\,\mathrm{km\,h^{-1}}$ on that section, the target speed was limited to $30\,\mathrm{km\,h^{-1}}$. This archieves a compromise between journey time and the peak power that has to be provided. Since on all other sections, the incline is actually negative for the loaded train, the required steady state power will always be lower than the $4\,\mathrm{MW}$ in **??** and the target speed was set to $45\,\mathrm{km\,h^{-1}}$. For the unloaded journey, the required traction power is lower, even on the steepest incline part, with the higher top speed. The target speed was therefore set to $65\,\mathrm{km\,h^{-1}}$ for all sections.

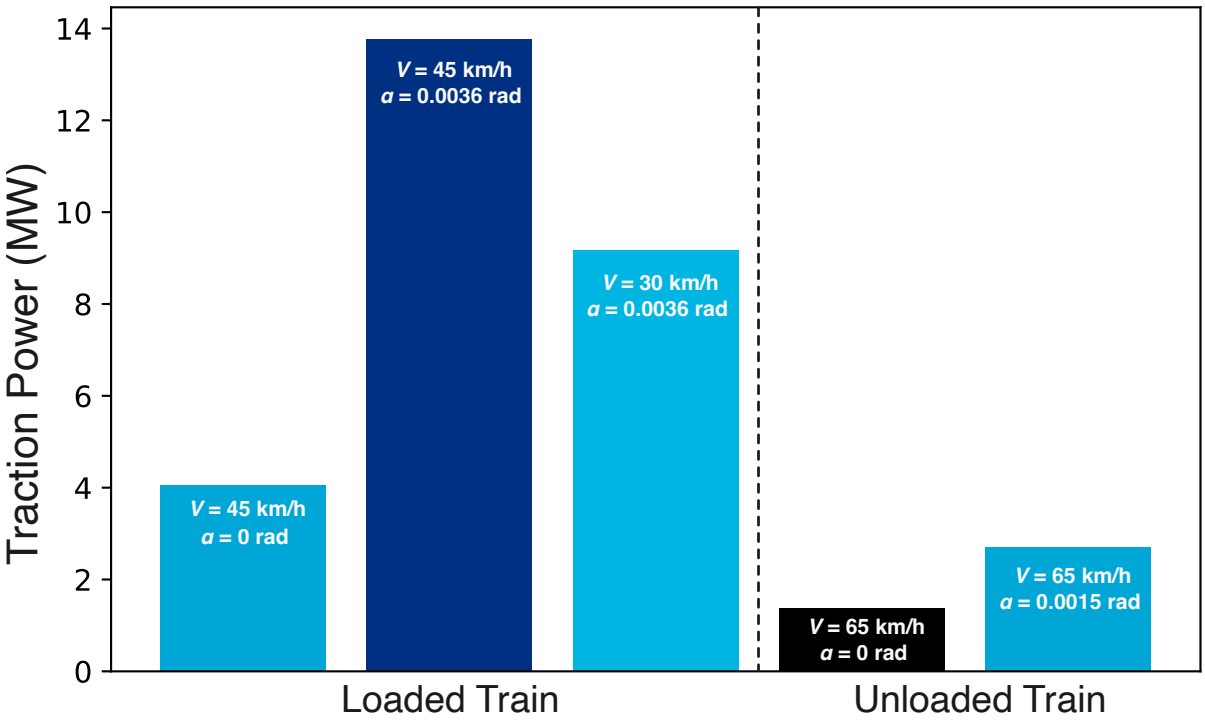

Figure 2.1: Steady state traction power for zero and maximum gradient for both, the unloaded and loaded train journey.

2.2 Train acceleration design

The acceleration of the train after a stop had the greatest freedom of design. By looking at other heavy-duty trains of similar size and weight, it could be concluded, that a maximum acceleration of $0.1\,\mathrm{m\,s^{-2}}$ is sufficient. Since the traction power required to archieve a specific acceleration in **??** increases with increasing speed, a constant acceleration is not a realistic model and would result in a spiking power for higher speeds. That is even more the case, since the resistive forces also increase with speed. To account for this, while still keeping a reasonably simple, quasi-static model, three different acceleration levels from stantstill up to the respective topspeed were chosen. The values for these acceleration levels were iteratively determined, by starting with educated guesses, that roughly keep $\frac{dV}{dt} \propto V^{-1}$, then computing the corresponding drive cycle for one acceleration period to top speed with a python skript, and finally calculating the traction power over time curve with the QSS vehicle model (see **??**). The acceleration levels were changed until the resulting traction power did not significantly exceed the maximum steady state power at maximum incline, to again avoid oversizing. The final values for the train acceleration as well as their cut off speeds are listed in **??**.

Table 2.2: Acceleration levels for the train starting from standstill and their corresponding cut off speeds, up until it is used.

$V_{cut,off}$ **(loaded) in** $\mathrm{m\,s^{-1}}$	15	30	-
$V_{cut,off}$ **(unloaded) in** $\mathrm{m\,s^{-1}}$	20	40	-
acceleration in $\mathrm{m\,s^{-2}}$	0.08	0.04	0.02

It has to be noted, that these values were designed for the loaded train on the stop with the least negative gradient. Since there is a stop on the loaded journey directly before the steep incline, the train also has to accelerate on that part. To keep the power withing limits, a very slow acceleration of $0.02\,\mathrm{m\,s^{-2}}$ was taken for the modelling of the loaded driving cycle after the first stop.

For the final values, one full acceleration period, as well as the corresponding traction power curve are shown in **??** after the stop in Inal. It can be observed, that the peak power stays below $10\,\mathrm{MW}$,

EV_Project - Mauretanian Train/plots/example_acceleration_period.png

Figure 2.2: Example of one acceleration period to loaded top speed of $45\,\mathrm{km\,h^{-1}}$.

due to the three acceleration levels. Note that an additional starting acceleration level, for speeds below $5\,\mathrm{km\,h^{-1}}$ is also present. This was added later in the design process, to limit the output power of the electric motor at very low speeds, where induction machines operate with a low efficiency. With the acceleration levels set, the final driving cycles could be created and imported into QSS together with the track slope that was extrapolated from the height map in **??**. They are shown in **??** for the loaded and unloaded journey.

EV_Project - Mauretanian Train/plots/drive_cycle.png

Figure 2.3: Final driving cycles for the loaded and unloaded journey.

3 Design of the Powertrain- Paul

Since the project takes into account not only the sole vehicle but also the charging infrastructure, the design of the powertrain was also factored in the design of the charging infrastructure.

3.1 Vehicle Power Train

Since the main motivation behind this project is to design a climate friendly alternative to the current diesel powered train, we decided to go with a fully electric drivetrain. This means that the train will be powered solely by electricity stored in batteries, which will be charged using renewable energy sources. This coice left us with two main optiosns for the drivetrain: a battery electric vehicle (BEV) or a fuel cell electric vehicle (FCEV). After careful consideration of the advantages and disadvantages of both options, we decided to go with a BEV drivetrain for our design. The main reasons fo this choice were logistical simplicity, efficiency, and safety.
Since hydrogen production always needs water, the production in the desert would be difficult. This said, the train would have to transport an enormous amount of hydrogen for the entire trip. This is already challenging since hydrogen has to be stored under high pressure. In the desert, such tanks would be exposed to high temperatures, which can increase the pressure inside the tanks and pose safety risks. Additionally, transporting hydrogen in such a harsh environment can be dangerous since the sand would be abrasive and could damage the tanks, leading to leaks.
Compared to this, the option of a BEV drivetrain in combination with charging islands along the track is much simpler. Electrical energy can be harvested from the abundant solar energy in the desert and stored in batteries off the train. This eliminates the need for transporting large amounts of fuel and the associated safety risks. Additionally, BEVs are generally more efficient than FCEVs, as they do not require the conversion of hydrogen into electricity, which can result in energy losses.

3.2 Charging Infrastructure

With the decision to go with a BEV drivetrain, the charging infrastructure also had to be designed accordingly. Charging the train directly from a 700 km long overhead line would be unfeasible due to multiple reasons. Not only would the installation of such a long overhead line be very expensive, but also maintenance would be difficult in the harsh desert environment. Additionally, since the harbour is the only stable grid conection on the area, the energy losses from transmitting electricity over such a long distance would be significant. Therefore, the decision was made to go with charging islands along the track. These charging islands will be equipped with large battery storage systems that can be charged using solar energy. The train can then stop at these islands to recharge its batteries during the journey.

3.3 Design Process

The necessary forces and energy requirements were determined through the design of the driving cycle (**??**). From there, the energy storage requirements were calculated by integrating power over time between stops. Initial rough estimations indicated a battery capacity of at least 6 to 8 MWh. While these early figures included significant safety margins to account for uncertainty, they were later refined using the QSS model, which incorporated efficiencies, mechanical losses, and auxiliary power. Motor design commenced based on the initial power estimations from the drive cycle, while the battery design was scaled accordingly as the calculations became more precise.

4 Motor and Transmission Design

To determine the characteristics of the train's power unit, we started from the power and force loads that the train would need to deliver to operate correctly. The power and force loads were calculated using these parameters, derived from the drive cycle and the physical characteristics of the train:

- Total mass of the train in the loaded and offloaded configurations
- Maximum speed of the train in the loaded and offloaded configurations
- Starting acceleration and average acceleration
- Wheel-rail friction coefficient
- Drag coefficient
- Inertia correction factor

Using these values, we derived the maximum power and maximum force to which the train would be subjected. Specifically, to calculate the train's maximum power, we selected the critical point where the loaded train, in the steepest area of the track, must start moving. Furthermore, we verified that this force could realistically be transmitted without slippage. For this test, we assumed an adherence coefficient in line with standard values in the railway industry. The results are shown below:

Table 4.1: Performance parameters of the iron ore train.

Peak Load Power	9.049 MW
Peak Load Force	2.245 MN
Traction Force Limit (0.46)	2.572 MN

These values, along with the transmission parameters, led to the creation of the electric motor's torque-speed curve. To haul the train, three locomotives, each equipped with six motors, were chosen. For the transmission, we referred to industry standards, i.e., a wheel diameter of 0.92 m, a gear ratio of 5, and a transmission efficiency of 97%. The operating parameters are therefore:

Table 4.2: Motor performance specifications.

Power	518.3 kW
Torque	11.829 kNm
Nominal Speed	418 rpm
Max Speed	3229 rpm

4.1 Power train design; evaluation of electrical parameters and efficiency map:

To implement the motor in QSS, it is necessary to define the power unit's electrical parameters and define an efficiency map for each operating point. To achieve these results, we chose an induction motor, as they are particularly popular for heavy-duty traction applications, are a consolidated technology, and require less maintenance than DC motors. We then chose to use the following circuit model to model the component:

5.png

Figure 4.1: Simplified electrical model of an Induction Machine

In particular, the following parameters are used to determine the following energy losses:

- Magnetization losses in iron, due to hysteresis phenomena and eddy currents: $P_{mag} \propto R_{fe}$
- Copper losses due to the Joule effect in the stator: $P_{cus} \propto R_s$
- Copper losses due to the Joule effect in the rotor: $P_{cur} \propto R_r/s$

We also considered mechanical losses due to friction and the engine's air cooling system: $P_{mech} = k_{att} \cdot \omega + k_{vent} \cdot \omega^3$[cite: 33].

4.2 Solver algorithm and final results

Once the parameters were chosen, an algorithm was developed that could solve the electrical circuit for each point of the Torque-Speed grid, thus providing an efficiency map. The Python solver for the script is shown below:

Code_1.png

Figure 4.2: Solver script part one

Code_2.png

Figure 4.3: Solver script part two

The algorithm was used iteratively to optimize the initial parameters based on the motor's operating range. Furthermore, the code allowed us to specify the maximum voltage and current required for the machine's correct operation. The results of this process are summarized in the images below:

eff map.png

Figure 4.4: Efficiency map

The numerical results of the algorithm are summarized below:

Calculated parameters

- **Nominal slip:** 1.65%
- **Nominal V_delta:** 686.28 V
- **Flux (Psi):** 2.5700 Wb

Peak Specifications

- **Line-to-Line Voltage (V_{rms}):** 1211.29 V
- **Phase Current (I_{rms}):** 356.74 A

- **Peak Efficiency:** 96.45%

Table 4.3: Efficiency in operating points.

Point	Torque [Nm]	Speed [rad/s]	Efficiency	Voltage [V]	Current [A]
Point 1	5720	90.60	95.54%	1210.15	284.79
Point 2	650	196.25	90.66%	1194.32	82.05

5 Energy Source

Before starting the design of our train's energy storage system, we re-confirmed the feasibility of our project. During our market research, we came across products such as the fully electric locomotives designed by FLXdrive, which reassured us that our project is viable [**?**].

5.1 Cell Chemistry

Starting on a cell level, different chemistries were considered. The main objectives are:

- **Long Lifecycle:** As this train will drive the whole day, every day. Any downtime directly results in a loss of income for every party involved.
- **Temperature Limitations**: During the day, temperatures can rise up to a stunning 50℃, whereas they might fall below 0℃ during a winter night.

Three different chemistries were considered: Lithium cobald oxide (LCO), Lithium iron phosphate (LFP) and Lithium-titanate battery (LTO). LCO batteries have a high power density and a somewhat short lifetime. Since the design is not optimizing for weight or volume, LCO batteries were discarded. LFP is broadly used for EVs and energy storage applications, even though its characteristic flat SOC curve makes it difficult to track the battery level. LTO is known for its extreme temperature endurance, fast charging capabilities, and unmatched lifespan. Even though LFP is cheaper to purchase, the longer lifecycle of LTO will make that chemistry cheaper in the long term than LFP. The longer lifetime (30,000 cycles), together with the better temperature performance, made us choose the Lithium Titanate Oxide Battery Cell - LTO 2.3V 40Ah cylindrical cell [**?**]. Note that for the charging islands, the cheaper LFP chemistry is chosen (see section **??**). Since there are multiple charging islands, the onboard battery will experience a factor of multiple cycles more over time than the charging island batteries. Table **??** provides an overview of the different cell chemistries considered.
The cylindrical cell was chosen, rather than the pouch cell, as it is a safer option in terms of robustness, and the weight reduction from the pouch cell is not part of our requirements.

Table 5.1: Comparison of considered battery cell chemistries.

Chemistry	Lifecycle (Cycles)	Operating Temp.	Relative Cost
Lithium Cobalt Oxide (LCO)	∼500 - 1,000	Poor	High
Lithium Iron Phosphate (LFP)	∼3,000 - 5,000	Moderate	Low
Lithium Titanate Oxide (LTO)	∼30,000	Excellent (-50℃ to 65℃)	High

5.2 Battery Configuration

The total onboard energy required is 12 MWh (**??**). This will be equally divided over the three locomotives, with each carrying a battery of 4 MWh. To reach the 800 V DC bus voltage, 348 cells will be placed in series:

$$V_{nominal} = N_S * V_{cell} = 348 * 2.3V = 800.4V \tag{2}$$

A total of 125 parallel connections will be made, resulting in a capacity of 5000 Ah:

$$C_{pack} = N_P * C_{cell} = 125 * 40Ah = 5000Ah \tag{3}$$

$$E_{pack} = V_{nominal} * C_{pack} = 4MWh \tag{4}$$

A single cell is estimated at the price of 39 euros [**?**]. The total amount of money for all 43,500 cells accumulates to 5 million euros. The SoC curves for both the loaded and the unloaded case are plotted in figure **??**, with data from the QSS simulation. It can be seen that the maximum SoC level will never exceed 93% and minium SoC will never go below 25% for the unloaded case and 9% for the loaded case.

The LTO cell can handle charge and discharge rates of 10C [**?**]. The peak charge current of the battery will occur during the acceleration part (see section **??**). A maximum of 6.2 kA will charge the onboard battery, or 1.24C, staying well below the current limit. Regenerative braking also won't deliver higher charge rates, since the downhill gradient is very shallow and the braking distance is very long (see figure **??** for the power curve). Discharging rates won't exceed 3000A, or 0.6C.

The total cell weight will be 54,375 kg per battery, with a volume of 30 m^3 (including terminals). The total weight of all three batteries is only 3.3% for the unloaded train and 0.7% for the loaded train. Therefore, the influence of this on the maximum acceleration, top speed and range can be neglected. Taking only the weight of the cells into account, the 4 MWh capacity per locomotive yields an energy-to-weight ratio of approximately 73.5 Wh/kg. Given the LTO cell's 10C discharge capability, the theoretical maximum power output is 40 MW per locomotive, resulting in a power-to-weight ratio of 736 W/kg. Regarding the placement, the battery packs will be housed centrally, occupying the space traditionally reserved for the diesel engine. The modules will be securely mounted in shock-resistant racks.

As this application integrates a BEV, the tank-to-wheel emissions are zero. Furthermore, since the electricity generation is completely decoupled from any grid and 100% of the energy is generated by solar power, the well-to-tank emissions are also zero. Thus, the expected well-to-wheel emissions are zero.

EV_Project - Mauretanian Train/plots/SOC_loaded.png

EV_Project - Mauretanian Train/plots/SOC_unloaded.pn

(a) Loaded case

(b) Unloaded case

Figure 5.1: SoC curves for both the unloaded and the loaded case.

To summarise, the train has three tractive locomotives, each carrying a battery with the following specifications:

- Energy: 4 MWh;
- Capacity: 5000 Ah;
- Chemistry: LTO;
- Nominal Voltage: 800 V;
- Number of cells in series: 348;
- Number of cells in parallel: 125;
- Total amount of cells: 43,500.

6 Power Conversion

Figure **??** shows an example of an electric locomotive without battery. The locomotives for the Mauritanian Train will look quite the same, besides carrying a battery and operating at different voltage levels.

6.1 Overall Architecture

A complete overview of the power transfer can be seen in figure .... For the current and power calculations, the maximum power is considered, as discussed in section ....
During standstill and acceleration, power is drawn from the 25 kV 1-phase AC overhead line. This 25 kV AC power is converted into 950 V AC before going onboard. This is done because the lower voltage is suitable for the traction equipment. Besides, 950 V AC is below the IEEE definition of high-voltage (1 kV AC) [**?**], saving difficulties around handling arc-flash risk, insulation coordination and creepage distances. The maximum total power is 36 MW used for both acceleration and charging at the same time. The transformer will convert this into
One-quadrant converters will convert the 950 V AC power into 950 V DC. According to the datasheet, the maximum DC voltage to charge the battery is 974.4 V [**?**].

6.2 Losses and Auxilary Power

EV_Project - Mauretanian Train//plots/Train_Power.png

Figure 6.1: Example of electric locomotive power transfer: overhead line (catenary) to traction motor [?]. This is for illustration purposes only, as this design integrates different components and voltage levels.

EV_Project - Marthetanthetanthetan

kinetic cooling. By manually constraining the train's charging speed to a minimum of 25 km/h, the pantograph continuously slides across "fresh," unheated wire, effectively spreading the friction and electrical arcing heat dynamically along the entire length of the track.

Electromagnetic Safety and Chassis Integrity

Transferring up to 20 MW of power wirelessly via induction generates massive stray magnetic fields. At this power scale, the magnetic flux would induce significant eddy currents in the train's steel chassis, the iron wheels, and the rails, effectively heating the locomotive structure like an induction stove and creating severe electrocution hazards. The overhead catenary system avoids these risks by isolating the electrical current within dedicated, shielded high-voltage cables leading directly to the onboard rectifiers, keeping the thermal load strictly contained within the train's dedicated active liquid-cooling loops.

Transmission Standard and Current Constraints

While a native low-voltage DC microgrid is highly efficient, the physical limits of the pantograph-to-wire contact point prohibit a purely DC catenary at this power scale due to the amperage limit. Standard DC railway voltages (e.g., 750 V DC to 3 kV DC) are practically limited to around 3 MW; attempting to push 20 MW at 3 kV DC would result in over $6,000$ Amps of current. This extreme amperage exceeds the physical limits of standard heavy-rail pantographs and would instantly melt the copper contact point. By specifying a 25 kV AC standard, the track can comfortably deliver 20 MW while keeping the current strictly under 800 Amps. This falls well within the safe operational limits of standard high-speed and heavy-freight rail pantographs. Consequently, the 25 kV AC overhead catenary is the selected architecture because it naturally manages the 1 MW thermal penalty through spatial distribution and passive cooling, avoids electromagnetic interference with the train chassis, and maintains a safe contact current. Furthermore, adopting this standard aligns the microgrid with proven global industry standards, ensuring commercial component availability, interoperability, and long-term mechanical reliability in the harsh Sahara environment.

7.2 Solar PV Generation and Environmental Adaptation

The energy generation subsystem serves as the primary power source for the off-grid charging islands. Due to the extreme environmental conditions of the Sahara Desert and the mechanical stresses inherent in heavy-haul rail operations, a highly specialized PV architecture is required to ensure long-term system reliability.

1. Mechanical and Environmental Placement Strategy

The PV arrays are intentionally set back 20–30 meters from the track-side rather than being placed between or directly adjacent to the rails. This design choice is necessitated by two primary factors: vibration mitigation and dust management. The $22,000$-tonne iron-ore trains generate significant ground-borne vibrations; consequently, direct rail-side placement would lead to accelerated cell micro-cracking and solder-joint failure. A 20 m setback provides sufficient soil-damping to protect the module's crystalline structure from these mechanical stresses. Furthermore, moving trains continuously shed iron-ore dust and lubricant aerosols, which are both highly opaque and electrically

conductive. By maintaining this setback, the system minimizes the volume of train-generated particulates reaching the array surface, thereby preventing severe shading losses and potential fire hazards associated with conductive dust buildup.

2. Module Technology: Heterojunction (HJT) and Bifaciality

Standard PERC (Passivated Emitter and Rear Cell) modules are rejected for this project due to their high temperature coefficients. Instead, the design specifies Bifacial HJT Modules to maximize yield in high-temperature environments. HJT cells possess a superior temperature coefficient of approximately $-0.25\%/°$C. In desert conditions where module temperatures frequently exceed $70°$C, HJT panels maintain significantly higher conversion efficiency compared to standard technologies that typically lose $\sim 15\%$ of their rated power under similar heat loads. Additionally, the Sahara sand provides a high albedo (reflectivity) of 30–45%. By utilizing bifacial modules, the system captures reflected irradiance on the rear surface, providing a "free" power boost of 15–20% without increasing the physical footprint of the array.

3. Autonomous Maintenance and Operations

In the desert environment, sand accumulation (soiling) represents the single largest cause of yield degradation, with the potential to reduce output by up to 30% if left uncleaned. To ensure consistent power delivery, the arrays are equipped with autonomous, waterless robotic cleaners. These robots perform nightly dry-sweeping cycles to remove sand and dust, ensuring the modules remain clear without consuming scarce local water resources or causing abrasive damage to the anti-reflective coatings.

4. Sizing and Auxiliary Load Management

To ensure the system remains self-sustaining, the total peak capacity (kW_p) of the array is oversized by 10–15% beyond the primary charging requirements of the locomotive. This overhead is dedicated to powering the Auxiliary Power Supply (APS). The APS drives the active liquid-cooling pumps and energy management systems required to keep the stationary LTO battery banks within their safe operational temperature window during peak daylight hours, ensuring the storage medium does not degrade under the intense ambient heat.

Table 7.1: Solar PV Engineering Design Summary

Design Element	Selection	Engineering Rationale
Placement	20–30m Setback	Vibration damping and iron-ore dust mitigation.
Cell Type	Heterojunction (HJT)	Low temp. coefficient (-0.25%/°C) for 50°C ambient.
Module Style	Bifacial	Utilizes 30–45% sand albedo for 15–20% yield boost.
Cleaning	Waterless Robotics	Prevents up to 30% yield loss from desert soiling.
Sizing	10–15% Oversize	Dedicated capacity for battery liquid-cooling loads.

7.3 Charging Island Energy Storage: Chemistry and Operational Parameters

Battery Chemistry Selection: Transition to LFP

Initial designs for the stationary off-grid charging islands considered Lithium Titanate (LTO) batteries due to their ability to sustain extreme discharge rates (up to 10C). However, a detailed load-profile

analysis of the updated 5-stop architecture revealed that the system's maximum discharge rates are significantly lower than originally anticipated.
Consequently, the station storage chemistry has been officially specified as **Lithium Iron Phosphate (LFP)**.

- **Engineering Rationale:** LFP offers a substantially higher volumetric energy density and a massively reduced capital cost per kWh compared to LTO. Modern liquid-cooled LFP cells can comfortably sustain continuous discharge rates of 1C to 2C, and short-duration bursts up to 3C. As demonstrated in the C-rate calculations below, the system never exceeds 2.2C, making LFP the optimal choice for both performance and project economics.

System C-Rate Verification Calculations

To validate the safety and longevity of the LFP cells, the maximum discharge C-rate ($C = \frac{\text{Power}}{\text{Capacity}}$) was calculated for both the DC stationary charging phase and the AC dynamic acceleration phase.

Scenario A: The DC Stationary Charging Phase (Max Output)
During the 30-minute cargo stop, the station utilizes the Tri-Port Megawatt Charging System (MCS) to push maximum power to the train.

- **Power Delivered to Train:** 11.25 MW (3 plugs $\times$ 3.75 MW)
- **DC-DC Converter Efficiency:** 95% ($\eta = 0.95$)
- **Actual Battery Discharge Power:** $\frac{11.25\ \text{MW}}{0.95} =$ **11.84 MW**
- **Worst-Case C-Rate (Stop 1):** Stop 1 has the smallest station battery at 5.4 MWh.
- **Calculation:** $\frac{11.84\ \text{MW}}{5.4\ \text{MWh}} =$ **2.19C**

Scenario B: The AC Dynamic Acceleration Phase (Max Output)
When the loaded train departs, it draws a massive 20 MW peak load from the overhead catenary to overcome static friction.

- **Power Delivered to Train:** 20.00 MW
- **AC Inverter/Transformer Efficiency:** 85% ($\eta = 0.85$)
- **Actual Battery Discharge Power:** $\frac{20.00\ \text{MW}}{0.85} =$ **23.53 MW**
- **Worst-Case C-Rate (Stop 4):** Stop 4 is the smallest station that provides AC power, with a 14.2 MWh battery.
- **Calculation:** $\frac{23.53\ \text{MW}}{14.2\ \text{MWh}} =$ **1.65C**

The absolute maximum discharge rate experienced by any station across the entire 700 km route is **2.19C** (lasting for a maximum of 30 minutes). Because this is well within the safe operational limits of LFP chemistry (which tolerates up to 3C bursts), the expensive LTO chemistry is unnecessary for the stationary islands. *(Note: The onboard train batteries will still utilize LTO due to strict weight limits requiring a smaller pack to absorb high continuous loads).*

Thermal Management in the Sahara Environment

While the electrical C-rates are fully compatible with LFP, the environmental conditions are not. LFP cells experience accelerated degradation and potential thermal runaway if operated continuously above $45°$C. Because ambient temperatures in the Mauritanian Sahara frequently reach $50°$C, passive cooling is mathematically insufficient.
To mitigate this, every LFP battery container at the charging islands will be equipped with an **Active Liquid Thermal Management System (LTMS)**. The physical solar arrays at each station have been specifically oversized by **15%** above the train's energy requirement. This dedicated energy buffer ensures the high-volume liquid cooling pumps and HVAC chillers can run at maximum capacity during the heat of the day without depleting the train's reserved charging energy.

7.4 Final Charging Station Sizing and Infrastructure Requirements

Sizing Methodology and Justifications

To determine the physical size of the solar arrays and battery storage at each of the 5 off-grid transit stops, the raw energy required by the train was adjusted to account for real-world thermodynamic and electrical losses. The final hardware sizing for each station is based on four sequential calculation steps:

- **Step 1: Converter Efficiencies (Station Grid Demand)**
  The system utilizes two different charging paths. The stationary DC Megawatt Charging System (MCS) is highly efficient, while the dynamic AC overhead catenary experiences higher conversion losses. To find the true energy the station must supply, we divide the train's demand by these efficiencies:
  $$\text{Grid Demand} = \left(\frac{\text{DC Energy}}{0.95}\right) + \left(\frac{\text{AC Energy}}{0.85}\right)$$
- **Step 2: The Sahara Thermal Penalty (Total Generation Required)**
  Operating in $50°$C ambient temperatures requires massive Active Liquid Thermal Management Systems (LTMS) to keep the batteries safe. A **15% energy penalty** is added to the Grid Demand specifically to power the liquid cooling pumps and HVAC chillers.
  $$\text{Total Generation} = \text{Grid Demand} \times 1.15$$
- **Step 3: Battery Health Buffer (Final Battery Size)**
  To maximize the lifespan of the Lithium Iron Phosphate (LFP) cells, the station batteries are sized so they never discharge completely to 0%. A **10% safety buffer** is added by dividing the required generation by 0.90.
  $$\text{Final Battery Size} = \frac{\text{Total Generation}}{0.90}$$
- **Step 4: Solar Array Sizing**
  The Sahara Desert provides excellent solar yield. The calculation assumes **6 Peak Sun Hours** per day and utilizes high-efficiency Bifacial HJT solar panels that generate **0.69 kW (690W)**

each, thanks to the extra light reflected off the desert sand (albedo effect).

$$\text{Number of Panels} = \frac{\text{Total Generation}}{6 \text{ hours} \times 0.69 \text{ kW}}$$

Final Hardware Configuration

Applying the methodology above to the simulated train energy demands yields the final infrastructure requirements shown in Table 1.

Table 7.2: Final Master Sizing Configuration (Efficiency-Adjusted)

Station	Energy Received (DC / AC)	Grid Demand	Total Gen. Req.	Final Battery Size	Final Solar Array Size
Stop 1	4,032 kWh / 0 kWh	4,244 kWh	4,881 kWh	**5.4 MWh**	**1,179 Panels** (0.8 MW)
Stop 2	5,622 kWh / 0 kWh	5,918 kWh	6,806 kWh	**7.6 MWh**	**1,644 Panels** (1.1 MW)
Stop 3	5,625 kWh / 5,975 kWh	12,951 kWh	14,893 kWh	**16.5 MWh**	**3,597 Panels** (2.5 MW)
Stop 4	5,625 kWh / 4,445 kWh	11,151 kWh	12,823 kWh	**14.2 MWh**	**3,097 Panels** (2.1 MW)
Stop 5	5,622 kWh / 0 kWh	5,918 kWh	6,806 kWh	**7.6 MWh**	**1,644 Panels** (1.1 MW)

Directional Load Variance (Loaded vs. Unloaded Journey)

It is critical to note that the charging stations were sized exclusively according to the "worst-case scenario" energy needs of the **westbound journey**, during which the train is fully loaded with 22,000 tonnes of iron ore.
When the train makes the **eastbound return journey** entirely unloaded, its energy demands drop significantly (peaking at a maximum of 5,622 kWh per stop). Because the unloaded train only requires the highly efficient DC stationary charging and never triggers the AC overhead lines, the infrastructure sized in Table 1 is more than sufficient to cover the return trip. No additional solar panels or batteries are required to support bi-directional travel.

Infrastructure Standardization Recommendation

As shown in Table **??**, **Stop 3 is the system bottleneck**, requiring nearly double the infrastructure of the other stations to support its 12.5 km dynamic AC charging zone.
Rather than custom-building five unique stations, it is highly recommended to standardize the construction into two classes to simplify the supply chain and maintenance:

- **Class A Stations (Stops 1, 2, and 5):** Standardized at ∼8.0 MWh of battery storage and ∼1.2 MW solar arrays.
- **Class B Stations (Stops 3 and 4):** Heavy-duty stations standardized at ∼16.5 MWh of battery storage and ∼2.5 MW solar arrays.

7.5 Operational Charging Times and Overhead Line Lengths

Calculation Methodology

To determine how long the train spends charging at each station, we use the maximum power transfer limits established in earlier sections:

- **DC Stationary Charging:** Maximum of **11.25 MW** (via three 3.75 MW MCS plugs).
- **AC Dynamic Charging:** Regulated to **20.0 MW** from the overhead catenary.

The time required (in minutes) is calculated using the standard power formula:

$$\text{Time (minutes)} = \left(\frac{\text{Energy (MWh)}}{\text{Power (MW)}}\right) \times 60$$

The Bottleneck Example (Stop 3 - Loaded Journey)

To demonstrate the system's capabilities, we analyze the most demanding segment of the route: Stop 3 on the loaded westbound journey, which requires a massive **11.60 MWh** of energy.

- **The DC Phase:** The train stops for its mandatory 30-minute cargo/logistics break. During this time, the MCS maxes out, delivering 5.625 MWh.
  - *Calculation:* $(5.625 \text{ MWh}/11.25 \text{ MW}) \times 60 =$ **30.0 minutes**.
- **The AC Phase:** The train departs, still needing 5.975 MWh to comfortably reach the next station. It raises its pantograph and draws 20.0 MW while accelerating.
  - *Calculation:* $(5.975 \text{ MWh}/20.0 \text{ MW}) \times 60 =$ **17.9 minutes**.
- **The Catenary Length:** Based on the train's acceleration profile, drawing power for 17.9 minutes requires exactly **12.5 km** of overhead AC wiring before the train can lower the pantograph and cruise on battery power.

The Zero-Delay Advantage (Unloaded Journey)

During the eastbound unloaded journey, the train is vastly lighter. The maximum energy required at any stop drops to 5.622 MWh.

- *Calculation:* $(5.622 \text{ MWh}/11.25 \text{ MW}) \times 60 =$ **30.0 minutes**.

Because the total energy demand can be entirely fulfilled within the standard 30-minute DC stationary window, the unloaded train **never uses the AC overhead lines**.

System-Wide Time and Distance Summary

The table below maps the exact charging times and physical AC infrastructure required for every physical station across both travel directions.
(Note: The unloaded stations have been reversed to match their physical location on the track. For example, Eastbound Stop 1 is physically located at Westbound Stop 5).

Table 7.3: Charging Durations and AC Infrastructure Requirements

Physical Station	Loaded DC Time	Loaded AC Time	Loaded AC Length	Unloaded DC Time	Unloaded AC Time	Unloaded AC Length
Stop 1	21.5 mins	0.0 mins	0.0 km	14.2 mins	0.0 mins	0.0 km
Stop 2	30.0 mins	0.0 mins	0.0 km	30.0 mins	0.0 mins	0.0 km
Stop 3	30.0 mins	17.9 mins	**12.5 km**	30.0 mins	0.0 mins	0.0 km
Stop 4	30.0 mins	13.3 mins	**9.0 km**	30.0 mins	0.0 mins	0.0 km
Stop 5	30.0 mins	0.0 mins	0.0 km	22.8 mins	0.0 mins	0.0 km

Logistical Conclusion

Because all DC charging occurs strictly within the pre-existing 30-minute operational stops, and all AC charging occurs dynamically while the train is physically driving, this electrification architecture introduces **zero minutes of schedule delay** to the mining company's logistics chain.

7.6 Power Electronics: Converters, Ratings, and Topology

Design Philosophy: Decentralized Power Electronics and Thermal Weight Reduction

A core principle of this hybrid rail architecture is maximizing ore payload by minimizing locomotive weight. While the locomotives must still carry Active Rectifiers to intake AC power from the dynamic overhead lines, the system shifts the heavy burden of stationary charging entirely to the off-board DC Megawatt Charging System (MCS).
If the train utilized AC catenary power for its 30-minute stationary cargo stops, the onboard rectifiers would require massive, heavy internal liquid-cooling systems to survive the $50°$C ambient heat at zero miles per hour (where there is no aerodynamic cooling). By delivering pure DC power directly to the batteries during stationary stops, the train physically bypasses its onboard AC converters. This allows the locomotives to carry significantly smaller, lighter AC rectifiers that are only activated during dynamic acceleration, relying heavily on natural high-speed airflow for thermal management.

Off-Board Converters (Station-Side)

The charging islands operate as isolated microgrids. They require three primary types of converters to manage the flow of energy from the solar panels to the batteries, and ultimately to the train.

1. **Solar Array Charge Controllers**
   - **Topology:** Unidirectional DC-DC Boost Converters with Maximum Power Point Tracking (MPPT).
   - **Power Rating:** Ranging from **0.8 MW to 2.5 MW** (scaled to match the specific Class A or Class B station array size).
   - **Function:** These converters step up the variable, low-voltage DC output of the Bifacial HJT solar panels to match the stable, higher voltage of the station's main DC bus (tied to the LFP battery bank).
2. **Stationary MCS Dispenser (The DC Fast Charger)**
   - **Topology:** Galvanically Isolated DC-DC Converters using a Dual Active Bridge (DAB) configuration.
   - **Power Rating: 11.25 MW Total** (Distributed as three separate 3.75 MW output channels).
   - **Function:** The DAB topology uses a high-frequency internal transformer to provide strict galvanic isolation between the station's microgrid and the train's chassis, which is a critical safety requirement for Megawatt charging. It precisely steps the station's DC voltage to the 800V required by the onboard train batteries.
3. **Dynamic Catenary Dispenser (The AC Grid-Former)**

- **Topology:** Modular Multilevel DC-AC Inverters (MMC) paired with a Phase-Shifting Step-Up Transformer.
- **Power Rating: 23.5 MW Peak** (designed to output a clean 20.0 MW after accounting for the 85% transmission efficiency).
- **Function:** Because the station batteries output DC, this massive inverter synthesizes the 25 kV single-phase AC required for the overhead catenary. The MMC topology is selected for its ability to handle extremely high voltages while producing a smooth, low-harmonic AC sine wave.

Station-Side Power Flow and Schematic Overview

Figure **??** illustrates the complete power flow logic for the off-grid charging islands. The system is designed around a central **800V Common DC Bus**, acting as the microgrid's backbone. The power flow is divided into four primary operational zones:

1. **Generation and Storage (The Backbone):**
   Energy enters the system via the Bifacial HJT Solar Arrays. The **C1 Charge Regulator** tracks the maximum power point of the solar panels and stabilizes the variable solar voltage onto the 800V DC bus. Connected directly to this bus is the **Stationary BESS** (the massive LFP battery bank), which acts as a bi-directional energy buffer—absorbing excess solar power during the day and discharging it when the train arrives.

2. **The Auxiliary Cooling Path:**
   Because the LFP batteries and power electronics must survive the $50°$C Sahara heat, 15% of the station's energy is strictly dedicated to the **LTMS Auxiliary Load** (Liquid Cooling and HVAC). Because large industrial cooling pumps require variable-speed AC motors, a dedicated **Auxiliary DC-AC Inverter (VFD)** draws DC power from the bus and converts it to variable-frequency AC to dynamically control the cooling speed based on the station's temperature.

3. **The Stationary Charging Path (DC Output):**
   When the train is parked for its 30-minute cargo stop, it utilizes the stationary charging path. Power flows from the DC bus through the **C2 Fast Charger**. This Dual Active Bridge (DAB) converter provides critical galvanic (physical) isolation for safety and ensures a perfectly stable 800V DC supply is delivered to the robotic MCS plugs connecting to the train.

4. **The Dynamic Acceleration Path (AC Output):**
   When the fully loaded train departs, it requires a massive 20 MW surge of power to overcome static friction. The station seamlessly routes power through the **C3 Dynamic Dispenser**, which acts as a grid-forming inverter. It converts the 800V DC into a clean AC waveform, which is then stepped up by a massive Phase-Shifting Transformer to standard **25 kV AC single-phase** power, feeding the overhead dynamic catenary lines.

7.7 Inter-operability

Physical Plug Standardization (MCS)

By adopting the Megawatt Charging System (MCS) rather than a bespoke railway connector, the 11.25 MW off-grid charging islands are not locked to the train. If the mining company later purchases

EV_Project - Mauretanian Train/plots/station_side_charging.png

Figure 7.1: Station-side Power Flow Schematic

battery-electric haul trucks or heavy earth-movers, those vehicles can drive up to the station and plug into the exact same standardized robotic MCS cables.

Overhead Transmission Standard

The overhead line is strictly regulated to 25 kV AC single-phase. Because this is the universal global standard for heavy-freight and high-speed rail [**?**], any commercial electric locomotive purchased in the future can seamlessly drive on this track infrastructure without modification.

Locomotive Modularity

Rather than designing a single 12 MWh mega-locomotive, the drivetrain uses three 4 MWh locomotives. This allows the mining company to easily swap units out for maintenance.

8 Future Developments

The project can be divided into two primary areas where future developments will have a significant impact: the charging islands and the train itself.

8.1 Impact on the Charging Islands

Currently, the charging islands are designed to produce solar energy exclusively for the train. However, as these islands are situated at stations within villages, the project could be expanded to provide energy to the entire community. This expansion could be transformative, as many rural areas still rely on fossil fuels for generators and heating or dont have access to energy at all [**?**]. Such a development aligns with the increasing global deployment of home energy storage systems [**?**], which would allow villagers to store excess energy and use it as needed.
A major advancement that would significantly increase storage efficiency is the development of batteries better suited for the extreme temperature ranges of the desert. This would reduce the auxiliary energy required for thermal management (cooling and heating), thereby increasing the overall efficiency of the system.

8.2 Impact on the Train

The same advancements in battery technology apply to the train. High-temperature-resilient batteries would allow the vehicle to operate more efficiently with a reduced energy load on its thermal management systems.
Additionally, the anticipated increase in power converter efficiency [**?**] represents a key development. Given that the train relies heavily on power electronics for its traction system, more efficient converters would have a substantial positive impact on the vehicle's total energy consumption.
Another potential advancement is the optimization of the DC bus voltage. Although the current 800 V architecture aligns with established electric vehicle standards, this specific application differs significantly from typical automotive use cases. Increasing the DC bus voltage to a higher level could improve overall efficiency by enabling more optimal motor operating points.

9 Simulation

EV_Project - Mauretanian Train/plots/tractive_force_vs_spee

Figure 9.1: Force-speed plot for the loaded and unloaded train with top speed at 0 and maximum gradient, as well as for the loaded train with the set acceleration levels at zero gradient.

EV_Project - Mauretanian Train/plots/efficiency_map_with_OP

Figure 9.2: Efficiency map of the induction machine motors used in the train with the operating points for the loaded journey

EV_Project - Mauretanian Train/plots/efficiency_map_with_OP

Figure 9.3: Efficiency map of the induction machine motors used in the train with the operating points for the unloaded journey

EV_Project - Mauretanian Train/plots/power_energy_plots.png

Figure 9.4: Total tractive torque, the machines have to provide during the loaded journey, as well as tractive and electrical power that have to be provided and the resulting electrical energy that has to be provided by the battery in between the charging stops.

EV_Project - Mauretanian Train/plots/power_energy_plots_unl

Figure 9.5: Total tractive torque, the machines have to provide during the loaded journey, as well as tractive and electrical power that have to be provided and the resulting electrical energy that has to be provided by the battery in between the charging stops.

10 Recap

EV_Project - Mauretanian Train/plots/Battery_model_ss.PNG

Figure 9.6: Battey model of the LTO Battery together with controller in QSS.

11 Peer Feedback

To evaluate our design process, we organized two peer review sessions scheduled one week apart. This timeline allowed us to implement the initial suggestions before the second session.

11.1 First Session

The first session was conducted with **Group 5**. We presented our design process and the current state of the project; at that stage, all individual subsystems had been designed, and we were focusing on system integration. Our primary objectives for the meeting were to ensure the completeness of our subsystems and to verify that our progress aligned with the project schedule.
The following points and questions were raised during the discussion:

- Are 25 kV overhead lines used with a single-phase or three-phase power supply?
- Is regenerative braking included in the design?
- How will the system handle sand and dust in a desert environment?
- Are there more cost-effective options than LTO (Lithium Titanate Oxide) batteries for the energy storage systems in the charging islands?
- Specific points of operation should be plotted on the motor's efficiency map.

EV_Project - Mauretanian Train/plots/battery_controller_ss.PNG

Figure 9.7: Custom battery controller, to ensure LTO battery gets not overcharged and simulation does not break.

While regenerative braking and environmental protection (sand/dust) were already integrated into our design, they had been omitted from the brief presentation. Following the session, we verified that 25 kV overhead lines are standardized for single-phase power supply [**?**]. Regarding the remaining feedback, we adjusted the charging island design to remove strict space constraints, allowing for the use of more economical battery types. Additionally, we overlaid the points of operation onto the motor efficiency map to provide better performance clarity.

12 Second Session

For the second session, the focus shifted from the system configuration toward the completeness and conciseness of the report. This second peer review was conducted with **Group 2** and **Group 7**. The first part of the discussion focused on clarifying the project's subsystems. The following points required further explanation:

- Why do we not use an overhead line for the entire length of the track?
- Is there a possibility of installing solar panels on the train?
- How did we determine the amount of solar radiation for our charging islands?

Regarding the first point, we had already considered this but had not included sufficient detail in the report; therefore, we refined **??** to clearly explain that decision. Solar panels on the train were

EV_Project - Mauretanian Train/plots/charging_model_ss.PNG

Figure 9.8: Charging model of the charging island infrastructure, per battery.

avoided to limit system complexity. Since the carriages are loaded from the top, any PV modules would need to be removable, introducing mechanical issues we preferred to avoid. As for the solar radiation, we utilized conservative assumptions. As stated in **??**, the six hours of peak radiation is a value intended to be matched daily [**?**].
Following these clarifications, the discussion shifted toward the report's structural details. Key feedback from both groups included:

- **Future Developments:** This section was missing at the time of review.
- **QSS Simulation:** Results plots were missing.
- **QSS Simulation:** Reasoning for the aerodynamic drag coefficient was needed.

The section for future developments (**??**) and the missing plots in **??** were added following the feedback. Finally, the aerodynamic drag coefficient was chosen based on an average from various articles. However, since **??** confirmed that aerodynamic forces are essentially negligible for this application, we did not go into further detail.